Composition and Elimination: Higher Derivatives, Determinants, and Schur Complements

N-20220409

§1.1 What remains when a variable is allowed to adjust?

Consider an energy with two real variables:

$$\Phi(x, z) = x^2 + xz + \frac{3}{2}z^2.$$

If we hold z fixed and allow x to find its cheapest position, completing the square gives

$$\Phi(x, z) = \left(x + \frac{z}{2}\right)^2 + \frac{5}{4}z^2.$$

The minimizing hidden coordinate is $x_*(z) = -z/2$, and the remaining energy is

$$g(z) = \min_x \Phi(x, z) = \frac{5}{4}z^2, \qquad g''(z) = \frac{5}{2}.$$

Why is the surviving curvature $5/2$ rather than the original second derivative $\Phi_{zz} = 3$? The hidden coordinate moves when z moves. Its response absorbs part of the energy change.

There are two ways to reach the answer, and they will meet in a useful theorem. Differentiating through the moving solution requires the chain rule and, at higher orders, careful management of derivatives of derivatives. Eliminating the hidden variable requires a little linear algebra; for a quadratic energy it produces a *Schur complement*. The equality between those computations is a precise geometric fact.

At the scalar level we can already see it:

$$\begin{pmatrix} \Phi_{xx} & \Phi_{xz} \\ \Phi_{zx} & \Phi_{zz} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \qquad \boxed{g'' = 3 - \frac{1^2}{2} = \frac{5}{2}.}$$

First we need to understand how derivatives compose. Then we can ask why eliminating variables changes the Hessian in exactly this way.

§1.2 Repeated derivations produce the higher Leibniz rule

Let D be a derivation on a commutative algebra, so

$$D(ab) = D(a)b + aD(b).$$

Applying D repeatedly gives

$$D^2(ab) = D^2(a)b + 2D(a)D(b) + aD^2(b),$$

and

$$D^3(ab) = D^3(a)b + 3D^2(a)D(b) + 3D(a)D^2(b) + aD^3(b).$$

The coefficients are not accidental.

Theorem 1.2.1 (Higher Leibniz rule)

For every integer $n \geq 0$,

$$D^n(ab) = \sum_{k=0}^{n} \binom{n}{k} D^k(a) D^{n-k}(b).$$

Proof. The result is clear for $n = 0$. Assume it holds for n. Differentiating once more gives

$$D^{n+1}(ab) = \sum_{k=0}^{n} \binom{n}{k} (D^{k+1}(a)D^{n-k}(b) + D^k(a)D^{n-k+1}(b)).$$

After shifting the index in the first sum, the coefficient of $D^k(a)D^{n+1-k}(b)$ becomes

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}.$$

This proves the induction step. □

There is also a combinatorial reading. During n differentiations, each derivative must land on either the first factor or the second. Choosing the k stages that land on a gives $\binom{n}{k}$. The formula is therefore a count of derivative-placement histories.

For a product of r factors, the same argument gives the multinomial rule

$$D^n(a_1 \cdots a_r) = \sum_{\alpha_1 + \cdots + \alpha_r = n} \frac{n!}{\alpha_1! \cdots \alpha_r!} \prod_{j=1}^{r} D^{\alpha_j}(a_j).$$

This form will reappear when differentiating a determinant row by row.

§1.3 Faà di Bruno records partitions of derivative labels

The ordinary chain rule has one outer derivative and one inner derivative:

$$(f \circ g)' = f'(g)g'.$$

At second and third order,

$$(f \circ g)'' = f''(g)(g')^2 + f'(g)g'',$$

$$(f \circ g)''' = f'''(g)(g')^3 + 3f''(g)g'g'' + f'(g)g'''.$$

The coefficients and monomials are controlled by set partitions. Think of the n derivative operations as carrying labels $1, \ldots, n$. Labels in the same block ultimately combine into one derivative of g; the number of blocks tells us which derivative of f appears.

For nonnegative integers $k_1, \ldots, k_n$, impose

$$k_1 + 2k_2 + \cdots + nk_n = n.$$

Here k_j is the number of blocks of size j. The total number of blocks is

$$m = k_1 + \cdots + k_n.$$

The number of set partitions with these block sizes is

$$\frac{n!}{k_1! \cdots k_n! (1!)^{k_1} \cdots (n!)^{k_n}}.$$

This gives the one-variable Faà di Bruno formula.

Theorem 1.3.1 (Faà di Bruno)

For sufficiently differentiable functions f and g,

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\substack{k_1,\ldots,k_n \geq 0 \\ k_1+2k_2+\cdots+nk_n=n}} \frac{n!}{k_1! \cdots k_n!} f^{(k_1+\cdots+k_n)}(g(x)) \prod_{j=1}^{n} \left(\frac{g^{(j)}(x)}{j!} \right)^{k_j}.$$

Proof. There is a short way to derive all the coefficients at once. Expand $g(x+h)$ around x and then expand f around $g(x)$, retaining only powers through h^n:

$$f(g(x+h)) \equiv \sum_{m=0}^{n} \frac{f^{(m)}(g(x))}{m!} \left(\sum_{j=1}^{n} \frac{g^{(j)}(x)}{j!} h^j \right)^m \pmod{h^{n+1}}.$$

To obtain h^n in the term with outer derivative $f^{(m)}$, choose k_j copies of the degree-j term. Necessarily

$$\sum_j k_j = m, \qquad \sum_j j k_j = n.$$

The multinomial coefficient for these choices is $m! / \prod_j k_j!$. It cancels the prefactor $1/m!$. Multiplying the coefficient of h^n by $n!$ gives exactly the stated formula. The blocks in the combinatorial proof are the same selections, with derivative labels restored. □

The same information is often compressed using the partial exponential Bell polynomials

$$B_{n,m}(x_1, \ldots, x_{n-m+1}) = \sum \frac{n!}{k_1! \cdots k_{n-m+1}!} \prod_{j=1}^{n-m+1} \left(\frac{x_j}{j!} \right)^{k_j},$$

where the sum satisfies

$$\sum_j k_j = m, \qquad \sum_j j k_j = n.$$

Then

$$(f \circ g)^{(n)} = \sum_{m=1}^{n} f^{(m)}(g) B_{n,m}(g', g'', \ldots, g^{(n-m+1)}).$$

Bell polynomials separate the combinatorics of composition from the particular functions being composed.

§1.4 A Fourth-Derivative Calculation

The integer partitions of 4 are

$$1+1+1+1, \qquad 2+1+1, \qquad 2+2, \qquad 3+1, \qquad 4.$$

They correspond respectively to the block-size data

$$(k_1,k_2,k_3,k_4) = (4,0,0,0),(2,1,0,0),(0,2,0,0),(1,0,1,0),(0,0,0,1).$$

Substitution into Faà di Bruno gives

$$\begin{aligned}(f\circ g)^{(4)} = {} & f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2g'' + 3f''(g)(g'')^2 \\ & + 4f''(g)g'g''' + f'(g)g^{(4)}.\end{aligned}$$

The coefficients $1,6,3,4,1$ count set partitions of the five displayed block types. For example, the type $2+1+1$ has

$$\frac{4!}{1!2!(2!)^1} = 6$$

partitions, while the type $2+2$ has

$$\frac{4!}{2!(2!)^2} = 3.$$

Take

$$h(x) = e^{\sin x}.$$

Since every derivative of $f(u) = e^u$ equals e^u,

$$\begin{aligned}h^{(4)}(x) = e^{\sin x}\big(& (\cos x)^4 + 6(\cos x)^2(-\sin x) + 3(-\sin x)^2 \\ & + 4(\cos x)(-\cos x) + \sin x\big).\end{aligned}$$

At $x=0$, this reduces to

$$h^{(4)}(0) = 1 - 4 = -3.$$

The Taylor series confirms the calculation:

$$e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + O(x^5),$$

so the fourth derivative at the origin is

$$4!\left(-\frac{1}{8}\right) = -3.$$

This check is useful because a missing block type in Faà di Bruno usually appears as an incorrect Taylor coefficient.

§1.5 Jets as Truncated Taylor Algebra

Leibniz notation becomes subtle beyond first order because the symbols $dx, d^2x, \ldots$ do not behave like ordinary powers of one number. The clean way to see what they encode is to work with Taylor polynomials modulo terms that are too small to matter at the chosen order.

Fix a point a. Two smooth functions have the same n-jet at a when their derivatives through order n agree there. The jet can be represented by the truncated Taylor polynomial

$$j_a^n f(h) = f(a) + f'(a)h + \frac{f''(a)}{2!}h^2 + \cdots + \frac{f^{(n)}(a)}{n!}h^n,$$

where calculations are performed modulo h^{n+1}. In other words, one works in the truncated polynomial algebra

$$\mathbb{R}[h]/(h^{n+1}).$$

Terms of degree greater than n are discarded because they cannot affect derivatives of order at most n.

This makes composition completely explicit. Suppose $g(a) = b$, and write the second-order jets as

$$j_a^2 g(h) = b + b_1 h + \frac{b_2}{2}h^2,$$
$$j_b^2 f(u) = c + c_1 u + \frac{c_2}{2}u^2.$$

Here

$$b_1 = g'(a), \qquad b_2 = g''(a), \qquad c_1 = f'(b), \qquad c_2 = f''(b).$$

Set

$$u = b_1 h + \frac{b_2}{2}h^2.$$

Since

$$u^2 = b_1^2 h^2 \pmod{h^3},$$

substitution gives

$$\begin{aligned} j_a^2(f \circ g)(h) &\equiv c + c_1 u + \frac{c_2}{2}u^2 \pmod{h^3} \\ &= c + c_1 b_1 h + \frac{c_2 b_1^2 + c_1 b_2}{2}h^2. \end{aligned}$$

Reading off coefficients recovers

$$(f \circ g)'(a) = f'(b)g'(a)$$

and

$$(f \circ g)''(a) = f''(b)(g'(a))^2 + f'(b)g''(a).$$

The second-order chain rule is therefore ordinary polynomial substitution followed by truncation.

At third order, write

$$j_a^3 g(h) = b + b_1 h + \frac{b_2}{2}h^2 + \frac{b_3}{6}h^3$$

and

$$j_b^3 f(u) = c + c_1 u + \frac{c_2}{2}u^2 + \frac{c_3}{6}u^3.$$

Modulo h^4,

$$u^2 = b_1^2 h^2 + b_1 b_2 h^3, \qquad u^3 = b_1^3 h^3.$$

Hence the cubic coefficient of the composition is

$$\frac{c_3 b_1^3 + 3c_2 b_1 b_2 + c_1 b_3}{6},$$

so

$$(f \circ g)'''(a) = f'''(b)(g'(a))^3 + 3f''(b)g'(a)g''(a) + f'(b)g'''(a).$$

The coefficient 3 is no longer mysterious: it comes from the two cross terms in u^2 together with the factorial normalization in the Taylor coefficients.

For a concrete check, take $f(u) = e^u$, $g(x) = \sin x$, and expand at 0. The relevant jets are

$$j_0^3 \sin(h) = h - \frac{h^3}{6}$$

and

$$j_0^3 e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6}.$$

Substitution gives

$$j_0^3(e^{\sin x})(h) = 1 + h + \frac{h^2}{2},$$

because the cubic contributions $-h^3/6$ and $+h^3/6$ cancel. Thus

$$(e^{\sin x})'''\big|_{x=0} = 0,$$

which agrees with the third-order chain rule.

The differential notation is a compressed version of this bookkeeping. If $y = f(x)$, then

$$dy = f'(x)\,dx$$

and

$$d^2y = f''(x)(dx)^2 + f'(x)d^2x.$$

When x itself depends on a parameter t, the first two pieces of its jet are

$$dx = x'(t)dt, \qquad d^2x = x''(t)(dt)^2.$$

Therefore

$$d^2y = (f''(x)(x')^2 + f'(x)x'')(dt)^2.$$

Only when x is chosen as an affine coordinate in the parameter does d^2x vanish. Under a nonlinear reparametrization it carries genuine second-order information. Jets explain precisely why treating d^2x as though it were merely $(dx)^2$ loses part of the chain rule.

§1.6 Differentiating Implicit Solutions

An equation may specify a function without ever solving for it. Suppose

$$F(x(z), z) = 0,$$

where x can itself be a vector and the square Jacobian F_x is invertible. The implicit function theorem gives a locally defined differentiable branch $x(z)$; differentiating the equation rather than explicitly solving it gives

$$F_x x' + F_z = 0, \qquad \boxed{x' = -F_x^{-1} F_z.}$$

For one real parameter z, differentiating again gives

$$F_x x'' + F_{xx}[x', x'] + 2F_{xz}x' + F_{zz} = 0,$$

hence

$$\boxed{x'' = -F_x^{-1}\left(F_{xx}[x', x'] + 2F_{xz}x' + F_{zz}\right).}$$

The bracket $F_{xx}[x', x']$ means the second derivative in the direction x' twice. The expression is exactly the second-order chain rule, with one extra linear solve. At still higher orders the terms are organized by the same partitions that appear in Faà di Bruno's formula.

For an example that is small enough to check to third order, let

$$F(x, z) = x + x^2 - z, \qquad x(0) = 0.$$

Then $(1 + 2x)x' = 1$, so $x'(0) = 1$. Another derivative gives

$$(1 + 2x)x'' + 2(x')^2 = 0, \qquad x''(0) = -2.$$

A third derivative gives $(1 + 2x)x''' + 6x'x'' = 0$, so $x'''(0) = 12$. The resulting jet is

$$\boxed{x(z) = z - z^2 + 2z^3 + O(z^4),}$$

which also follows by expanding the positive branch

$$x(z) = \frac{-1 + \sqrt{1 + 4z}}{2}.$$

The implicit calculation scales to high-dimensional problems because each derivative order asks us to solve with the same linearized operator F_x. This will be especially helpful when the equation expresses stationarity of an energy.

§1.7 Parametric Differentiation

Let a regular plane curve be given by

$$x = x(t), \qquad y = y(t),$$

with $x'(t) \neq 0$. Differentiation with respect to x means applying

$$\frac{d}{dx} = \frac{1}{x'(t)}\frac{d}{dt}.$$

Hence

$$\frac{dy}{dx} = \frac{y'}{x'},$$

and

$$\frac{d^2y}{dx^2} = \frac{x'y'' - y'x''}{(x')^3}.$$

The numerator is a determinant:

$$x'y'' - y'x'' = \det\begin{pmatrix} x' & y' \\ x'' & y'' \end{pmatrix}.$$

Applying the operator once more gives

$$\frac{d^3y}{dx^3} = \frac{(x')^2y''' - 3x'x''y'' + \left(3(x'')^2 - x'x'''\right)y'}{(x')^5}.$$

The growing expression is another composition problem: one repeatedly composes multiplication by $1/x'$ with d/dt.

For the semicubical parabola

$$x = t^2, \qquad y = t^3, \qquad t > 0,$$

we obtain

$$\frac{dy}{dx} = \frac{3t}{2}, \qquad \frac{d^2y}{dx^2} = \frac{3}{4t}, \qquad \frac{d^3y}{dx^3} = -\frac{3}{8t^3}.$$

Since $y = x^{3/2}$ on this branch, direct differentiation gives exactly the same values. The singular behavior as $t \to 0$ records the failure of x to be a valid local coordinate at the cusp.

§1.8 Determinants inherit a multinomial Leibniz rule

Write an $m \times m$ matrix-valued function by rows:

$$A(t) = \begin{pmatrix} r_1(t) \\ \vdots \\ r_m(t) \end{pmatrix}.$$

Because the determinant is multilinear in its rows,

$$\frac{d}{dt} \det A(t) = \sum_{i=1}^{m} \det(r_1, \dots, r_i', \dots, r_m).$$

Repeated differentiation and the multinomial Leibniz rule give

$$\frac{d^n}{dt^n} \det A(t) = \sum_{\alpha_1 + \cdots + \alpha_m = n} \frac{n!}{\alpha_1! \cdots \alpha_m!} \det(r_1^{(\alpha_1)}, \dots, r_m^{(\alpha_m)}).$$

Thus the higher derivative of a determinant is organized by distributing n derivative operations among m rows.

At first order there is also a basis-independent formula. Let $\operatorname{adj}(A)$ denote the adjugate matrix. Cofactor expansion gives

$$\frac{d}{dt} \det A(t) = \operatorname{tr}\big(\operatorname{adj}(A(t)) A'(t)\big).$$

If $A(t)$ is invertible, then

$$\operatorname{adj}(A) = \det(A) A^{-1},$$

so

$$\boxed{\frac{d}{dt} \det A(t) = \det A(t) \operatorname{tr}\big(A(t)^{-1} A'(t)\big).}$$

This is Jacobi's formula. Equivalently,

$$\frac{d}{dt} \log \det A(t) = \operatorname{tr}\big(A(t)^{-1} A'(t)\big)$$

whenever the logarithm is defined.

For

$$A(t) = \begin{pmatrix} e^t & t \\ 0 & 1 + t \end{pmatrix},$$

we have

$$\det A(t) = e^t(1+t),$$

and therefore

$$\frac{d}{dt}\det A(t) = e^t(2+t).$$

Jacobi's formula gives the same result because

$$\operatorname{tr}(A^{-1}A') = 1 + \frac{1}{1+t}.$$

Multiplication by $\det A = e^t(1+t)$ yields $e^t(2+t)$.

§1.9 Matrix Products as Composition

Let

$$B : U \to V, \qquad A : V \to W$$

be linear maps. After bases are chosen, their composition is represented by the product AB. Its entries are

$$(AB)_{ij} = \sum_k A_{ik}B_{kj}.$$

The index k is internal: it labels the intermediate coordinate in V, and summation removes it from the final answer. This is a contraction between the input index of A and the output index of B.

Associativity follows because finite sums may be regrouped:

$$\begin{aligned}((AB)C)_{ij} &= \sum_k \left(\sum_\ell A_{i\ell}B_{\ell k}\right) C_{kj} \\ &= \sum_\ell A_{i\ell}\left(\sum_k B_{\ell k}C_{kj}\right) \\ &= (A(BC))_{ij}.\end{aligned}$$

Commutativity fails because reversing the order reverses the composition. Even when both products exist, AB and BA generally represent maps on different spaces or different operations on the same space.

Block multiplication is the same rule with each scalar entry replaced by a compatible linear map. For example,

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} x \\ z \end{pmatrix} = \begin{pmatrix} Ax + Bz \\ Cx + Dz \end{pmatrix}.$$

This notation becomes powerful when one block of variables is to be eliminated.

§1.10 Schur complements are exact block elimination

Consider

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A is invertible. Multiplying on the left by a block elimination matrix gives

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & S \end{pmatrix},$$

where

$$\boxed{S = D - CA^{-1}B}$$

is the Schur complement of A in M.

This identity immediately gives

$$\det M = \det A \, \det S.$$

It also gives a block solve. For

$$Ax + Bz = b, \qquad Cx + Dz = c,$$

the first equation gives

$$x = A^{-1}(b - Bz).$$

Substitution into the second equation produces the reduced system

$$Sz = c - CA^{-1}b.$$

After solving for z, one recovers x by back-substitution. The Schur complement is exactly what remains after the variable x has been eliminated.

§1.11 A complete block-elimination example

Take

$$M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad D = (3).$$

Since

$$A^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

we obtain

$$CA^{-1}B = \frac{2}{3}, \qquad S = 3 - \frac{2}{3} = \frac{7}{3}.$$

Therefore

$$\det M = \det A \, S = 3 \cdot \frac{7}{3} = 7.$$

Now solve

$$M \begin{pmatrix} x_1 \\ x_2 \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The reduced right-hand side is

$$2 - CA^{-1}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2 - \frac{2}{3} = \frac{4}{3}.$$

Hence

$$\frac{7}{3}z = \frac{4}{3}, \qquad z = \frac{4}{7}.$$

Back-substitution gives

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A^{-1}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\frac{4}{7}\right) = A^{-1}\begin{pmatrix} 3/7 \\ 0 \end{pmatrix} = \begin{pmatrix} 2/7 \\ -1/7 \end{pmatrix}.$$

A direct multiplication verifies

$$M\begin{pmatrix} 2/7 \\ -1/7 \\ 4/7 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}.$$

The calculation shows concretely that the reduced scalar $7/3$ contains the full effect of the eliminated two-dimensional block.

§1.12 Block inverse and positive definiteness

If both A and S are invertible, block factorization gives

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}.$$

The formula is best remembered as elimination followed by back-substitution, not as four unrelated blocks.

For a symmetric block matrix

$$M = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}$$

with A positive definite, completing the square gives

$$\begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix} = (x + A^{-1}Bz)^T A(x + A^{-1}Bz) + z^T(D - B^TA^{-1}B)z.$$

Consequently,

$$M \succ 0 \quad \Longleftrightarrow \quad A \succ 0 \text{ and } D - B^TA^{-1}B \succ 0.$$

Thus the same Schur complement controls elimination, determinants, inverses, and positivity. In statistics it also appears as a conditional covariance; in constrained optimization it appears after eliminating primal or dual variables.

§1.13 Optimized Curvature and Schur Complements

Let $\Phi(x, z)$ be a twice continuously differentiable real function, with $x \in \mathbb{R}^p$ and $z \in \mathbb{R}^q$. Suppose at (x_0, z_0) we have

$$\nabla_x \Phi(x_0, z_0) = 0, \qquad A = \nabla^2_{xx}\Phi(x_0, z_0) \succ 0.$$

The implicit function theorem gives a local stationary branch $x_*(z)$ through x_0, and positive definiteness makes each nearby point a strict local minimum in the x-directions. Put

$$g(z) = \Phi(x_*(z), z)$$

and write the full Hessian, at a point of this branch, as

$$\nabla^2\Phi = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}, \qquad B = \nabla^2_{xz}\Phi.$$

Theorem 1.13.1 (Hessian after eliminating stationary variables)
At every point of the branch where A remains positive definite,

$$\boxed{\nabla^2 g(z) = D - B^T A^{-1} B.}$$

Proof. Differentiate the stationarity equation $\nabla_x\Phi(x_*(z), z) = 0$:

$$A\, D_z x_* + B = 0, \qquad D_z x_* = -A^{-1}B.$$

By the ordinary chain rule,

$$D_z g = (\nabla_x\Phi)^T D_z x_* + \nabla_z\Phi = \nabla_z\Phi.$$

This is the envelope theorem in its simplest form: the first-order movement of the minimizer makes no contribution because the x-gradient vanishes there. Differentiate once more:

$$\nabla^2 g = D + B^T D_z x_* = D - B^T A^{-1} B.$$

The Schur complement is therefore exactly the Hessian of the reduced energy. □

In particular $B^T A^{-1} B$ is positive semidefinite, so $\nabla^2 g \preceq D$: when a hidden variable is allowed to relax, the visible energy cannot become *stiffer* at second order than it was with that hidden coordinate held fixed.

Return to the two-variable energy from the beginning. With $A = 2$, $B = 1$, $D = 3$, the formula gives $g'' = 5/2$ without needing to complete a square. For the three-dimensional matrix of the block-elimination example, take

$$\Phi(x, z) = \frac{1}{2}\begin{pmatrix} x \\ z \end{pmatrix}^T \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix}, \qquad x \in \mathbb{R}^2.$$

The minimizing response is

$$x_*(z) = -A^{-1}Bz = \begin{pmatrix} -2/3 \\ 1/3 \end{pmatrix} z.$$

The effective energy becomes

$$g(z) = \frac{1}{2}\left(3 - \frac{2}{3}\right) z^2 = \frac{7}{6} z^2.$$

Its second derivative is $7/3$, precisely the Schur complement that previously appeared in the determinant and block solve. This is the promised meeting of implicit differentiation, Hessian geometry, and elimination.

§1.14 A Gaussian ellipse makes the elimination visible

The same quadratic calculation explains a familiar probability picture. Let

$$M = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix} \succ 0, \qquad p(x,z) \propto \exp\left[-\frac{1}{2}\begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix}\right].$$

The matrix M is the *precision*, or inverse covariance, of a centered Gaussian distribution. Completing the square gives

$$\begin{pmatrix} x \\ z \end{pmatrix}^T M \begin{pmatrix} x \\ z \end{pmatrix} = (x + A^{-1}Bz)^T A (x + A^{-1}Bz) + z^T (D - B^T A^{-1} B) z.$$

For a fixed z, the conditional Gaussian $x \mid z$ is centered at $-A^{-1}Bz$ and has covariance A^{-1}. Integrating out x instead gives

$$\int_{\mathbb{R}^p} e^{-\frac{1}{2}(x+A^{-1}Bz)^T A (x+A^{-1}Bz)}\, dx = \frac{(2\pi)^{p/2}}{\sqrt{\det A}},$$

which does not depend on z. Therefore the *marginal* law of z has precision

$$\boxed{S = D - B^T A^{-1} B}$$

and covariance S^{-1}. Conditional behavior and marginal behavior are different, yet both are read off from the same completed square.

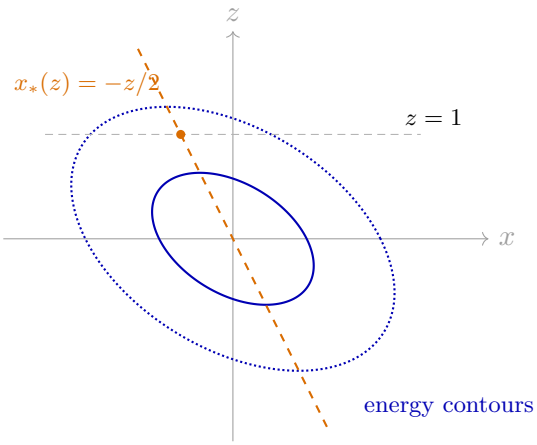


Figure 1.1: Exact contours of the joint Gaussian energy $\Phi(x,z) = x^2 + xz + \frac{3}{2}z^2$, at levels $\Phi = 1/2$ and $\Phi = 2$. Each horizontal slice has its lowest energy at $x = -z/2$ (orange line); eliminating x leaves the quadratic profile $g(z) = 5z^2/4$.

For this particular matrix, $x \mid z$ has mean $-z/2$ and variance $1/2$, whereas the marginal z has variance $2/5$. The ellipse encodes both statements. It also explains why eliminating hidden Gaussian variables is an exact operation, unlike approximating an arbitrary nonlinear energy by a quadratic.

§1.15 Differentiating through Elimination

The derivative rule for matrix inverses follows immediately from differentiating $A(t)A(t)^{-1} = I$:

$$\boxed{\frac{d}{dt}A^{-1} = -A^{-1}A'A^{-1}.}$$

If the blocks of $M(t)$ vary differentiably, then $S(t) = D - CA^{-1}B$ satisfies

$$\begin{aligned} S' = D' - C'A^{-1}B - CA^{-1}B' \\ + CA^{-1}A'A^{-1}B. \end{aligned}$$

Thus one can differentiate a *reduced* system without ever constructing a full matrix inverse. The same inverse derivative gives a compact second-order identity for any positive definite matrix curve:

$$\boxed{\frac{d^2}{dt^2}\log\det A = \operatorname{tr}\left(A^{-1}A'' - A^{-1}A'A^{-1}A'\right).}$$

Indeed, the first derivative is $\operatorname{tr}(A^{-1}A')$ by Jacobi's formula. The product rule and the inverse derivative produce the two displayed terms. When M, A, S are all positive definite,

$$\log\det M = \log\det A + \log\det S,$$

so differentiating the small Schur complement also differentiates the contribution of the eliminated variables to the log-determinant. These are standard block-elimination identities; an accessible general reference is Boyd and Vandenberghe, *Convex Optimization*.

The most satisfying link is now an exact one. Taylor jets and Faà di Bruno describe how a composed response changes; the implicit-function theorem finds the response of variables determined by equations; and block Gaussian elimination removes those variables. For an optimized quadratic problem the two routes agree in the single matrix $D - B^TA^{-1}B$. That matrix is simultaneously an effective Hessian, a reduced linear system, and the precision of a Gaussian marginal. The combinatorics of higher derivatives still has its own interest, but there is now a concrete problem in which it and linear algebra genuinely work together.